Hall Ticket Number:

MR-22

Code: B424CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Database Management Systems

(Common to CSE, IT, CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Explain Entity set with suitable example.	1	2	1	1,2
1. b)	What are the most common errors you can potentially face in data modeling?	1	1	1	1,2
1. c)	List the constraints in relation model.	1	1	2	1,2
1. d)	Describe different operators in relational algebra.	1	2	2	1,2
1. e)	Discuss different types of SQL operators.	1	2	3	1,2,3,12
1. f)	Illustrate UNION and INTERSECT in SQL.	1	2	3	1,2,3,12
1. g)	Name the disadvantages of Time stamp protocol.	1	1	4	1,2,3,12
1. h)	Classify the Lock based protocol.	1	2	4	1,2,3
1. i)	Explain memory storage device hierarchy.	1	2	5	1,2
1. j)	Mention the objectives of file organization.	1	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Outline Database – System Applications in detail.	5	2	1	1,2
2. b)	What are five main functions of a Database administrator?	5	1	1	1,2
	OR				
2. c)	Compare Weak Entity Set and Strong Entity Set with example.	5	2	1	1,2
2. d)	Design an ER diagram for a Hospital management system that includes entities such as patients, employees, doctors, nurses, room, receptionist, test report, and billing. Specify the relationships between these entities and their attributes.	5	5	1	1,2
3. a)	Describe constraints on a single relationship.	5	2	2	1,2,3,12
3. b)	Explain Tuple Relation Calculus with example.	5	2	2	1,2,3,12
	OR				
3. c)	Write the given queries in SQL student(Roll-number, Name, Address, Age, Phone) i. Select names of all students who stay in Delhi. ii. Find the name, Roll-number of all students in ascending order by age. iii. Select Name, Address, AVG (age) from student based on Address.	5	4	2	1,2,3,12
3. d)	i. Develop Salesman Table (salesperson ID, name, city and commission). ii. Create a view to Update the city name which salesperson ID is '5007'.'.	5	4	2	1,2,3,12
4. a)	Write the given queries in SQL Sailors(sid: integer, sname: string, rating: integer, age: real); Boats(bid: integer, bname: string, color: string); Reserves(sid: integer, bid: integer, day: date). i. Find the ids and names of sailors who have reserved two different boats on the same day.	5	4	3	1,2,3,12

P.T.O.

	ii. Find the ids of sailors who have reserved a red boat or a green boat. iii. Find the name and the age of the youngest sailor.				
4. b)	Mention the differences between triggers and stored procedures with example.	5	3	3	1,2,3,12
	OR				
4. c)	Discuss about Third and BCNF Normal form with example.	5	3	3	1,2,3,12
4. d)	Discuss the problems caused by decomposition in Schema refinement.	5	2	3	1,2,3,12
5. a)	Explain Timestamp based protocols in detail.	5	5	4	1,2,3
5. b)	Describe Multiple Granularity in detail.	5	2	4	1,2,3
	OR				
5. c)	What is the Distinction between a Deferred and an Immediate Update?	5	3	4	1,2,3
5. d)	Describe recovery algorithm using log records from transaction failure.	5	2	4	1,2,3
6. a)	Differentiate Static Hashing and Dynamic Hashing.	5	2	5	1,2,3
6. b)	Compare Ordered indexing and Hashing.	5	2	5	1,2,3
	OR				
6. c)	Explain Clustered Index and Non-Clustered Index with example.	5	2	5	1,2,3
6. d)	Illustrate different Types of File Organizations techniques.	5	2	5	1,2,3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome